

# Continuous FACS Sorting of Double Emulsion Picoreactors with a 3D-Printed Vertical Mixer

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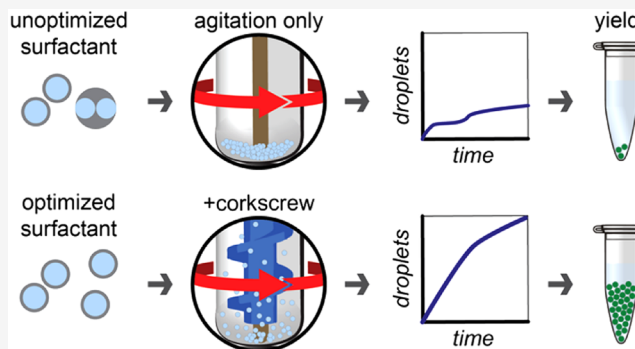


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**ABSTRACT:** High-throughput screening and directed evolution using microfluidic picoreactors have produced high-activity enzymes. In this approach, a substrate is coencapsulated with a candidate enzyme, and individual picoreactors are sorted based on an activity reporter. While many approaches use water-in-oil droplets (single emulsions) for fluorescence-activated droplet sorting (FADS) on custom-fabricated microfluidic devices that require integrated optics and electronics, recent approaches have lowered the engineering barriers to adoption by using simple microfluidic droplet generators to produce water-in-oil-in-water droplets (double emulsion picoreactors, DEs) that can be sorted with commercial FACS (fluorescence-activated cell sorting). Despite the simplified engineering requirements, high variability in loading rates and low yields during loading are barriers to efficient DE FACS sorting. Here, we optimized surfactants to enhance DE stability and demonstrated that a 3D-printed corkscrew on the sample line acts as a vertical mixer to enable more continuous loading. With these optimized loading conditions, we analyzed 1.17 million DEs in four 10 min sorting rounds with a mean frequency of 480 Hz (390 Hz including sample exchanges); in a mock sort of 10% fluorescent DEs, we achieved  $89 \pm 1\%$  accuracy and  $78.0 \pm 0.9\%$  recovery with our optimized loading protocol. Overall, improved ease of use and throughput for FACS-sortable DEs should expand the accessibility of directed evolution in controlled in vitro environments.



## INTRODUCTION

Library selection for directed evolution has produced many feats of macromolecular engineering, particularly in the production of novel enzymes and biocatalysts.<sup>1</sup> Cell-based assays<sup>2–8</sup> (e.g., phage display, yeast display, bacterial FACS, etc.) are convenient because they produce protein in situ and link genotypes and phenotypes; however, they are impractical for many engineering applications because they are incompatible with extreme reaction conditions (e.g., labile substrates, pH, temperature, toxic compounds, secreted compounds). Microfluidic droplet picoreactors offer a powerful alternative and support reaction conditions incompatible with cell-based screens.<sup>9</sup> These picoreactors physically link a target macromolecule with a functional readout (e.g., an assay product) through encapsulation, reduce reagent use (1–100 pL per reaction), and are compatible with convenient readouts (e.g., fluorescence,<sup>10–12</sup> mass,<sup>13</sup> absorbance,<sup>14,15</sup> or electrochemistry<sup>16</sup>). Microfluidic generators also allow for customizable workflows and highly controlled mixing and picoinjection.

While single emulsion (water-in-oil) picoreactors have been successfully used for selection via fluorescence-activated droplet sorting (FADS), this approach requires custom microfluidic sorters integrating flow control, optics, and

piezoelectrics. To increase accessibility, recent approaches (Table S1) have developed double emulsion picoreactors (DEs, water-in-oil-in-water) that can be sorted with commercial FACS machines (fluorescence-activated cell sorting). FACS offers multicolor sorting (up to 18 fluorescence channels)<sup>17</sup> and high peak sorting rates (up to 20 kHz), and FACS-sortable DEs have been used for multiple assays, including RT-PCR,<sup>18</sup> aptamer sensors for enzymatic products,<sup>19</sup> vesicle rupture,<sup>20</sup> and fluorogenic enzyme substrates.<sup>9,21–24</sup> The diameter of sorted particles must be <30% of the size of the nozzle to prevent DE shearing during sorting<sup>9,19,25–27</sup> and larger nozzles have slower maximum sort rates.<sup>26,28</sup>

For DEs, a trade-off exists between throughput and control. Heterogeneous bulk-generated DEs are generally <20  $\mu\text{m}$  in diameter and can be sorted with smaller, faster nozzles, but low

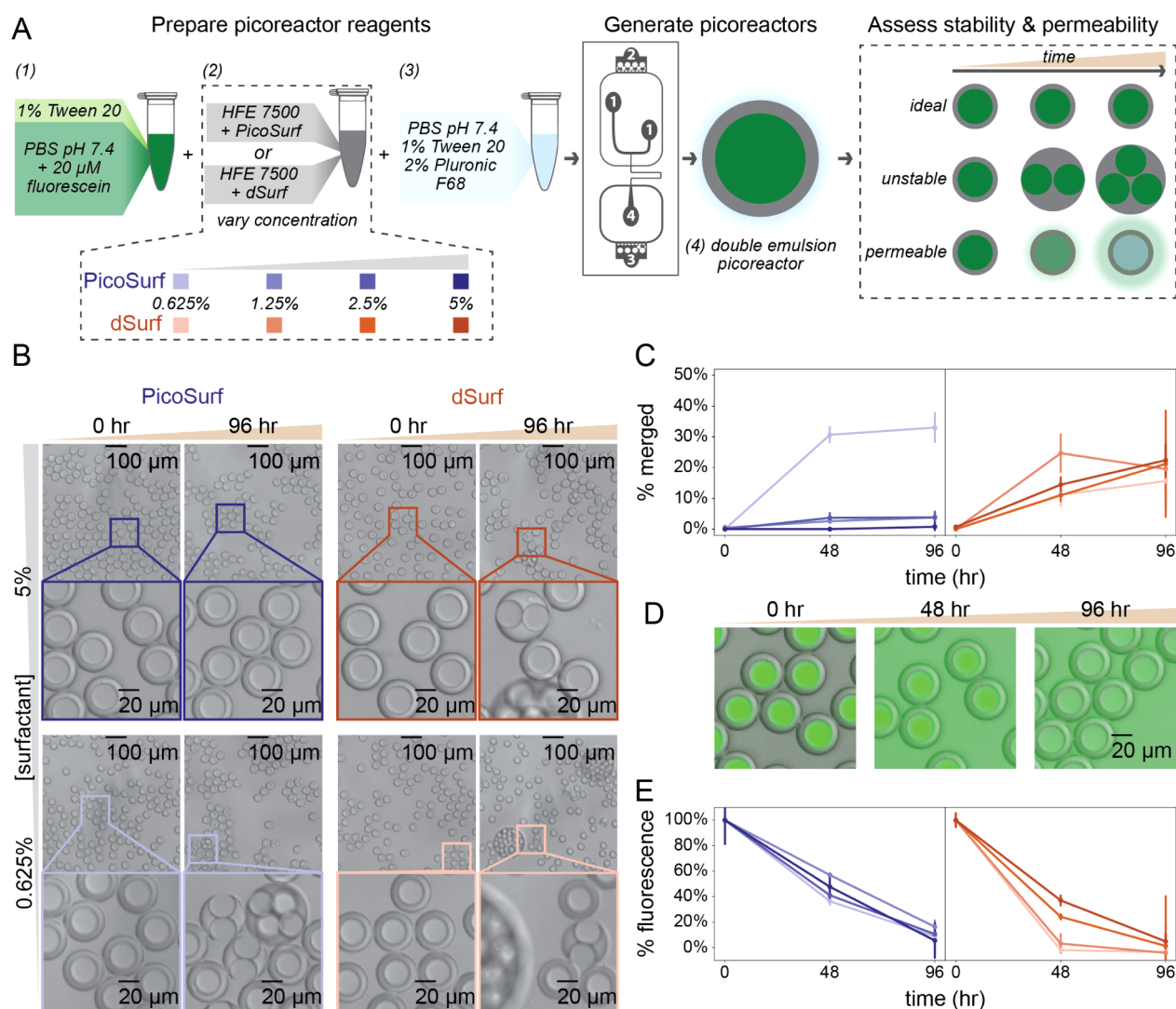
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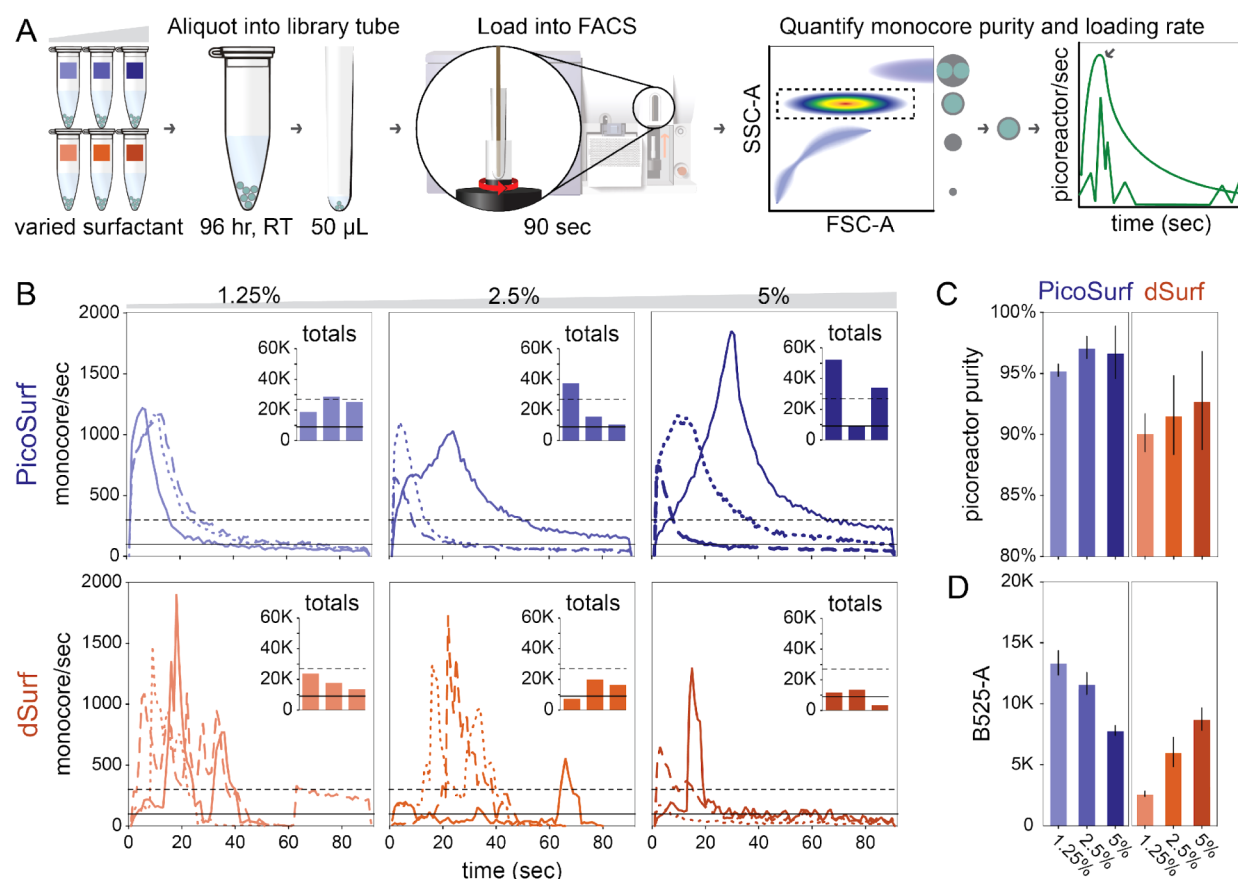
**Figure 1.** Optimize oil surfactants for double emulsion picoreactor stability. A) Workflow for generating DE picoreactors and analyzing stability and permeability by microscopy. Color gradients indicate varying concentrations of PicoSurf (blue) and DSurf (orange); numbered inputs and outputs match ports/channels on the picoreactor generator schematic. B) Brightfield microscopy images of picoreactors at 0 and 96 h (brown gradient) for 5% and 0.625% surfactant concentrations (gray gradient). Scale bar: 100  $\mu$ m. Inset images show 5 $\times$  zoom of (outline) one section of each respective image. Inset scale bar: 20  $\mu$ m. C) Percentage of aqueous cores in merged picoreactors as a function of surfactant conditions and time. Error bars represent the standard deviation across three images. D) Merged brightfield and green fluorescence microscopy images for 5% PicoSurf picoreactors at 0, 48, and 96 h. Scale bar: 20  $\mu$ m. E) Retained initial fluorescence as a function of surfactant conditions and time. Error bars represent standard deviation across three images.

signal-to-noise ratios and high variability in reaction volume size and reagent concentration impede many applications, and applications such as reverse transcription from encapsulated mammalian cells require larger DEs.<sup>18,27</sup> Reflecting a greater desire for control over DE size variation and the composition of reagents in the DE core, the field has shifted to microfluidic DEs (Figure S1A) that are monodisperse<sup>26,29–31</sup> and compatible with mammalian cells;<sup>18,27</sup> however, these DEs tend to be large (30–50  $\mu$ m) and therefore require 130  $\mu$ m nozzles to maintain stable breakoff of the sorter stream, which limits peak reported sorting rates to 2–3 kHz (Figure S1B).

Despite reportedly high *peak* sorting rates, maintaining these rates over long time frames is technically challenging. As a result, the vast majority of DE studies in the past decade analyzed <250 000 events and did not evolve a new macromolecule (Table S1 and Figure S1C,D). These sorting challenges have been further exacerbated by the increasing use

of dense fluorocarbon oils compatible with a wide variety of biochemical and cellular reactions, as large, dense DEs settle rapidly in the sample tube. As DEs settle, they become inaccessible to the loading tube, and the loading rate decays rapidly, limiting throughput until the sample is unloaded and manually resuspended. Additionally, DEs are more fragile than cells because larger particles are subject to more intense shearing forces, and DEs lack a cytoskeleton. Sheared DEs frequently clog the FACS sample lines and nozzle, further reducing the overall sorting efficiency.<sup>26,28</sup>

Here, we systematically optimized DE loading rates and stability with the goal of sorting 1 million DEs per hour (approximately 300 Hz). In this optimization, we varied surfactants and surfactant concentrations and attached a 3D-printed corkscrew to the FACS loading tubes to prevent DE settling. Using our optimized conditions, we demonstrated the ability to consistently sort at peak loading rates of 500–1000



**Figure 2.** FACS analysis to optimize the oil surfactant for picoreactor loading. **A**) Workflow for analyzing monocore DE picoreactor FACS loading rates as a function of surfactant conditions. Example diagram (far right) depicts inconsistent and consistent (arrow) loading. **B**) Monocore DE picoreactor loading rates over three independent 90 s runs (solid line, long dashes, and short dashes). Data are color-coded by surfactant conditions as in Figure 1. Black lines indicate benchmark rates: 100 (solid) or 300 (dashed) picoreactors/sec. Inset bar charts indicate total monocore picoreactors accumulated for each run. **C**) Picoreactor purity (percentage of monocore picoreactors) out of all FACS scattering events. Error bars represent standard deviation over 3 independent runs. **D**) Monocore picoreactor permeability as the mean ratio of green fluorescence (B525) over FSC-A. Error bars represent standard deviation over three independent runs.

Hz over several minutes with  $89 \pm 1\%$  accuracy and  $78.0 \pm 0.9\%$  recovery. With 4 rounds of iterative sample loading, we successfully sorted  $10^6$   $40 \mu\text{m}$  DE picoreactors in  $<1$  h, including sample exchange times. In future work, we anticipate that these technical advances will significantly extend the duration of uninterrupted sorting and enable routine analysis of  $10^5$ – $10^6$  member libraries in directed evolution campaigns.

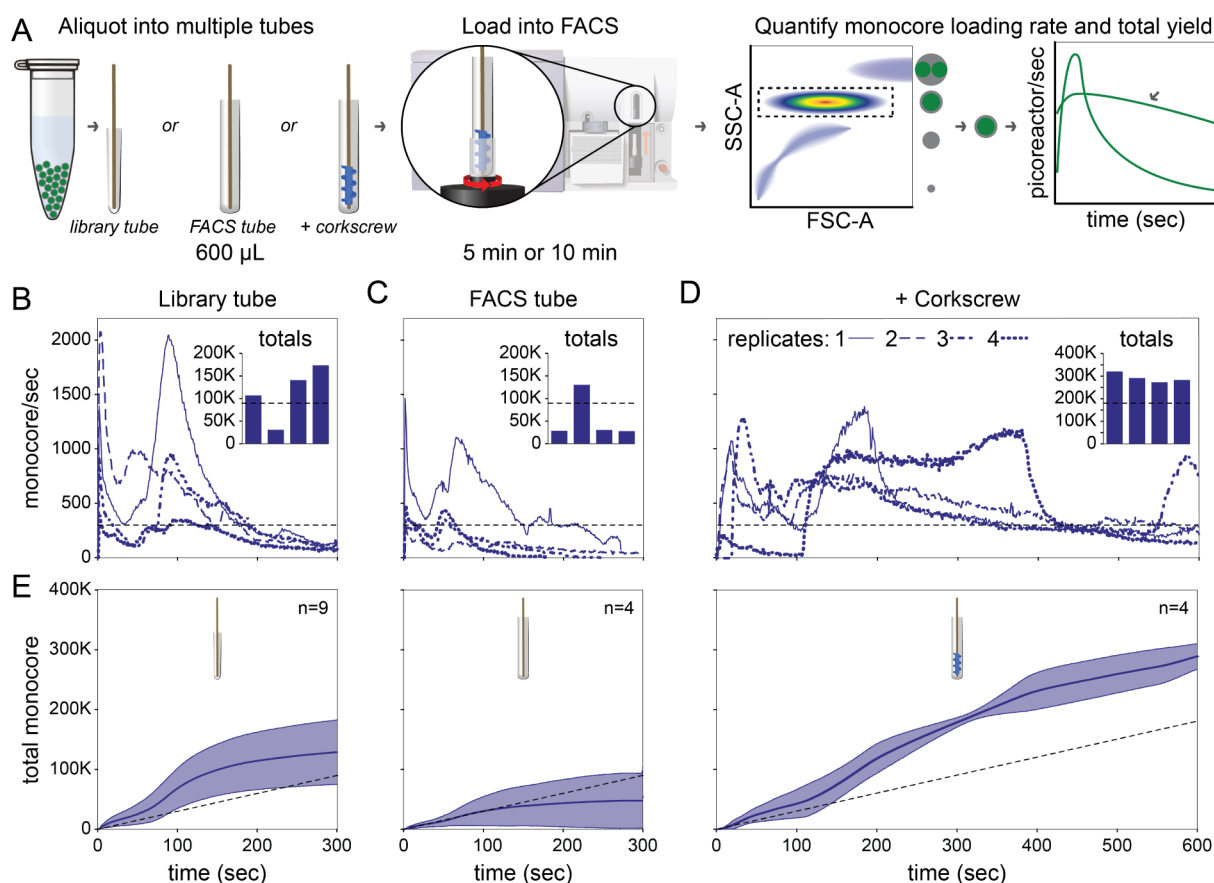
## RESULTS

**Optimizing Oil Surfactants for Double Emulsion Picoreactor Stability.** DEs are composed of an inner aqueous layer, a fluorocarbon (e.g., HFE7500), or hydrocarbon (e.g., mineral oil) oil shell, and an outer aqueous layer that allow DEs to be suspended within the FACS sheath fluid (Figure 1A). The added surfactants migrate to each water/oil interface, where they lower interfacial surface tension to promote DE formation and prevent coalescence. Some commercially available nonionic droplet surfactants (e.g., dSurf and PicoSurf) have been optimized for improved compatibility with cellular and biochemical assays relative to ionic surfactants and hydrocarbon oils;<sup>9,32,33</sup> however, their impact on long-term DE stability and sorting remains uncharacterized.

To test the impacts of these surfactants on DE stability, we: (1) generated DEs with HFE7500 oil and 0.625–5% (w/v) of

either PicoSurf or dSurf, (2) added  $20 \mu\text{M}$  fluorescein to the inner aqueous phase to label the aqueous core, and (3) monitored DE merging with optical microscopy after 0, 48, and 96 h ( $n = 437$ – $1255$  DE cores per time point; Figures 1B,C and S2 and Table S2). For moderate and higher concentrations of PicoSurf, DEs showed little unwanted merging even after 96 h ( $<1\%$  merged for 5% PicoSurf). In contrast, dSurf DEs showed substantial merging that continually increased over time (15–25% merged for all dSurf concentrations after 96 h, and nearly half of the DEs merged in 96 h for the lowest concentration of PicoSurf). In addition, merged PicoSurf DEs typically contained 2 aqueous cores, while dSurf DEs formed large multicore aggregates (Figure 1B). Overall, these results indicate that PicoSurf concentrations  $\geq 1.25\%$  are optimal for long-term DE stability.

Droplet crosstalk is a practical concern because small molecules diffuse into hydrophobic environments.<sup>34–39</sup> Surfactant properties can also impact the rate of solute leakage from DEs.<sup>40–43</sup> As fluorescein is particularly challenging to encapsulate without leakage, it can sensitively report surfactant-dependent differences in the transport of small molecules between droplets. To test whether conditions that enhance stability alter leakage, we also imaged DEs to quantify the fluorescence of the inner aqueous core during the time course (Figure 1D). Over 96 h, the background-subtracted

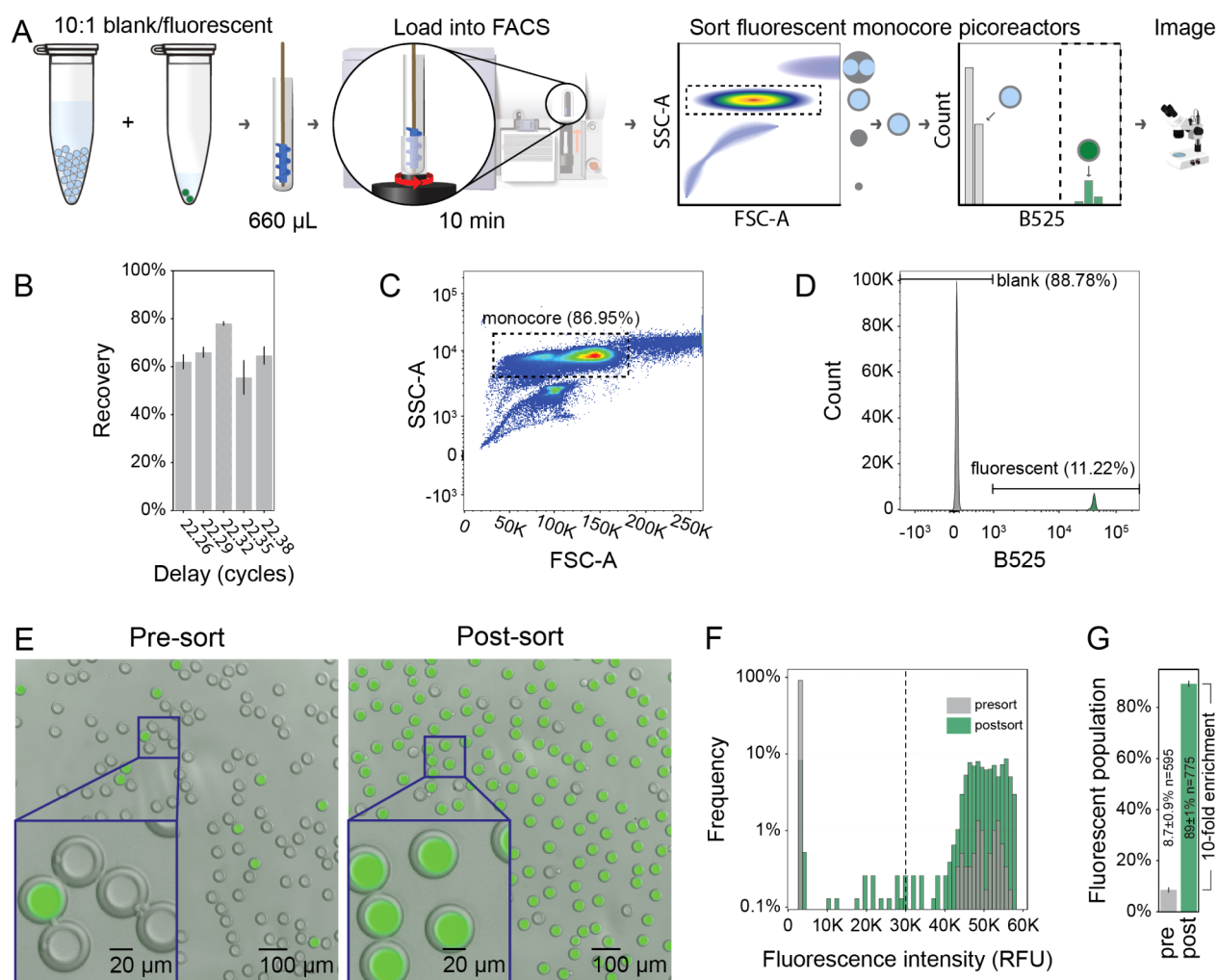


**Figure 3.** FACS analysis to increase sample agitation for improved picoreactor loading. A) Workflow for analyzing 5% Picosurf monocore DE picoreactor FACS loading rates across sample tubes. Example diagram (far right) depicts unsustained and continuous (arrow) loading. B–D) Monocore picoreactor loading rates over four independent runs (solid line, long dashes, short dashes, and alternating dashes). Runtimes are 5 min in B (library tube) and C (FACS tube) or 10 min in D (FACS tube with corkscrew). Dashed black lines indicate a benchmark rate of 300 picoreactors/sec. Inset bar charts indicate total monocore picoreactors accumulated for each run. E) Mean total accumulated picoreactors as a function of run time. *n*-values indicate the number of independent FACS runs for each loading condition. Error bars represent standard deviation over all *n* runs; dashed black lines indicate total picoreactors at a benchmark rate of 300 picoreactors/sec.

fluorescein intensity within the inner core decreased to 0–20%. For PicoSurf DEs, the raw fluorescence signal for the DE cores and the background fluorescence asymptotically approached the same value over 96 h (Figure S3), consistent with an equilibration between the core and the outer solution on the order of days. Similar multiday equilibration kinetics have previously been observed with both single and double emulsions.<sup>41,44</sup> Fluorescein leakage was faster for dSurf DEs, reaching equilibrium around 48 h. The final fluorescence intensity was generally higher for PicoSurf DEs at the 96 h time point (Figures 1E and S3 and Table S3). Fluorescence leakage was slower for aqueous cores in merged DEs, likely due to a reduced surface area-to-volume ratio (Figure S4). Overall, PicoSurf DEs appear to retain small molecules more effectively than dSurf.

**Picosurf-Containing Picoreactors Exhibit More Consistent Sorting Rates.** Successfully characterizing large DE libraries requires not only that DEs remain stable but also that they can be loaded into the FACS machine at consistent rates over time. To test the impact of surfactant conditions on DE loading and fluorescence retention, we used our previously generated double emulsions (Figure 1) with 20  $\mu$ M fluorescein and 1.25%, 2.5%, or 5% of either PicoSurf or dSurf, incubated for 96 h to match the imaging in Figure 1, aliquoted into 50  $\mu$ L (65 000–75 000 DEs) of each sample into a small-volume

library tube, loaded samples onto a BD FACS Aria III, and quantified for monocore purity and loading rate over time (Figure 2A); 3 replicates of each sample were loaded under 300 rpm agitation for 90 s. For many samples, DE events did not appear for the first 1–2 min of loading; in these cases, we started recording data after event rates first reached 20–40 events/s (Figures 2B, S5, and S6). Overall, PicoSurf DEs exhibited a more consistent loading pattern: loading rates increased over 10–20 s, peaked between 500 and 2000 Hz, and quickly dropped below 300 Hz (often dropping below 100 Hz by the end of the run) (Figure 2B). In contrast, dSurf DEs exhibited multiple brief loading rate spikes and frequent periods below 100 Hz (Figure 2B). The median total monocore DE yield after 90 s of loading was significantly higher for PicoSurf samples relative to dSurf, with 5% PicoSurf DEs yielding a median of  $30\,000 \pm 20\,000$  DEs over 90 s (equivalent to a constant integrated sorting rate of 375 Hz) (Table S4). Excess sample remained in all cases, such that the total yield is not comparable to the total generated DEs. Consistent with the high stability that we observed for PicoSurf DEs from microscopy analysis (Figure 1), monocore purity was higher for PicoSurf (Figure 2C and Table S5) than for dSurf and exceeded 95%. To analyze DE permeability, we analyzed the fluorescein fluorescence intensity (488/525 nm laser excitation and emission). PicoSurf DEs had a higher mean



aliquots of 5% PicoSurf DEs (approximately 10 min of DE generation or 750 000–850 000 DEs) and sorted for at least 5 min. Peak rates using library tubes and the larger sample volume ranged from 300 to 2000 Hz, with total DE yields of 50 000–175 000 (Figures 3B and S8 and Table S7); as with prior sorts, excess sample remained in all cases. Peak rates with FACS tubes ranged from 300 to 1500 Hz, but sorting rates and total DE yield were consistently lower than with library tubes (yield ranged from 20 000 to 120 000 DEs) (Figures 3C and S9 and Table S8). Peak rates with the corkscrew and FACS tubes ranged between 1000 and 1500 Hz and were more continuous than in conditions without the corkscrew, making it possible to extend the sort time to 10 min (Figures 3D and S10 and Table S9). Total DE yields for the longer sort times ranged from 270 000 to 325 000 and were more consistent than under conditions without the corkscrew. The FACS tube alone decreased loading rates, but adding the corkscrew resulted in more reliable peak yields and more continuous loading rates.

We next examined which loading conditions would meet our benchmark of sorting  $10^6$  DEs in 1 h (equivalent to sustained 300 Hz loading, Figure 3E). Both the library tube and FACS tube + corkscrew conditions had mean sorting rates that were consistently higher than 300 Hz. With library tubes, we analyzed  $1.16 \times 10^6$  DEs over nine 5 min runs; using FACS tubes with the corkscrew, we analyzed  $1.17 \times 10^6$  DEs over four 10 min runs, equivalent to a mean frequency of 480 Hz. Even allowing for 2 min of sample handling during each sample exchange, this would maintain a mean frequency of 390 Hz. Mean sorting rates noticeably decreased after 5 min, likely due to either sample settling or sample depletion. Additional representative commercial surfactants (5% Xdrop Surf, 5% Ran 008-FluoroSurf, and 2.2% ionic Krytox FSH; see Table S1) also allowed generation of stable DEs under our standard generation conditions (Figure S11). DEs generated with the nonionic Xdrop and ionic Krytox surfactants could be loaded into the FACS sorter at rates above our target rate and comparable to those for PicoSurf DEs, but DEs with the Ran surfactant showed much less consistent loading rates that generally remained below our target rate (Figures S12 and S13). These results demonstrate that surfactant optimization results in an efficient workflow for sorting double emulsion reactors and that mechanically resuspending the pellet of dense DEs allows for more continuous sorting protocols.

**FACS Sorting with Corkscrew-Assisted Loading Preserves Picoreactor Integrity.** Screening libraries in DEs further requires accurate recovery of the desired populations via FACS sorting (without contamination from unwanted DEs). To demonstrate that the addition of the corkscrew did not compromise DE integrity or sorting, we attempted to select only fluorescent DEs from a population of 10:1 blank-to-fluorescent DEs (approximately 660  $\mu$ L) using the FACS tube + corkscrew loading condition (Figure 4A). To ensure robust linkage between detected and sorted events, we first manually calibrated the FACS drop delay by counting the recovery of 50 DEs sorted onto a glass slide, yielding a recovery rate of  $78 \pm 1\%$  at the optimal delay setting of 22.32 cycles (Figure 4B). Using this optimal delay setting, we then detected 354 041 FACS events (Figures 4C and S14) including 303 833 monocore DEs (86.9%); consistent with the expected loaded ratio, 11.2% of monocore DEs were fluorescent (MFI >  $10^3$ ; Figure 4D).

To test the ability to successfully recover only desired DEs, we sorted 30 000 fluorescent DEs and analyzed the pre- and postsort populations with fluorescence microscopy (Figure 4E). While the unsorted sample remaining in the FACS tube showed signs of mechanical damage (Figure S15), the sorted population consisted predominantly of intact monocore DEs. Before and after sorting,  $9 \pm 1\%$  ( $n = 595$ ) and  $89 \pm 1\%$  ( $n = 775$ ) of DEs were fluorescent, respectively. This selection represented a 10-fold enrichment (Figure 4F). These results demonstrate that DEs loaded with increased agitation can be efficiently sorted by FACS and recovered intact.

## DISCUSSION

No published papers using the relatively large and monodisperse DEs preferred for modern molecular and cell biology applications have reported analyzing more than 250 K DEs in total – with one exception, where DEs were osmotically shrunk before sorting.<sup>29</sup> This limit likely stems from technical challenges associated with FACS loading that prevent consistent sorting of large DE libraries. To address this, we systematically examined how different surfactant conditions and sample loading methods impact FACS sorting throughput for large (40  $\mu$ m) microfluidic DEs. DEs with 5% PicoSurf were stable for multiple days and could be consistently sorted via a FACS machine over 10 min runs when loaded using a 3D-printed corkscrew and standard FACS tubes. Using these optimized conditions, we analyzed 1.17 million DEs in four 10 min sorting runs with sustained sorting rates of 500–1000 Hz and  $89 \pm 1\%$  sort accuracy, laying the foundation for future work screening large DE libraries.

Parallelization over multiple FACS sorters (standard for single-cell workflows) is an obvious next step in increasing throughput. As an alternative, further optimization of DE size and loading methods could enable sorting of  $10^7$ – $10^8$  large, monodisperse DEs per day using a single sorter. Published work has demonstrated generation of 15  $\mu$ m<sup>24</sup> and 20–23  $\mu$ m<sup>19,29</sup> monodisperse DEs; as the maximum sort rate is inversely correlated with DE size, these >23  $\mu$ m DEs have been sorted at peak rates of 6–8 Hz. For assays compatible with DEs of this size, maintaining these higher peak rates would boost throughput an additional 20-fold relative to this work. In addition, the incorporation of other mixer geometries (e.g., vertical ribbon mixers that produce poloidal mixing currents<sup>45</sup> or the use of density-matched oils<sup>46</sup> or outer fluids<sup>47</sup>) could maintain DE suspension for longer times, boosting ease of sorting by increasing the duration of hands-off sort times. By using commercial sorters<sup>48–50</sup> to process larger DEs (50–500  $\mu$ m), future work could open new selection assays for organoids, filamentous fungi, and bacterial communities and enable lower signal-to-noise detection methods such as absorption or morphology.

Here, we optimized DE stability and loading using common biochemical assay buffers (e.g., PBS). While many droplet generation conditions, surfactants, and FACS parameters can be used across a wide variety of assays,<sup>9,18,25,27,30</sup> chemically similar surfactants can yield unexpectedly variable results (e.g., nonionic PEG triblock copolymers in Figures 2 and S13). For optimizing DE sorting for specific applications, we provide a more detailed discussion of practical considerations for successful sorting in the **Extended Methods** in **Supporting Information**. Overall, continual end-to-end optimization of FACS-sortable DEs will develop this technology into a

modular plug-and-play platform compatible with a diverse range of assays and encapsulated inputs.

## METHODS

Extended methods are available in the [Supporting Information](#) and on the Open Science Framework (OSF) at <https://osf.io/jzyve/>.

**Picoreactor Generation.** Double emulsion picoreactors were generated from custom-fabricated PDMS devices using previously reported designs.<sup>9,18,25,27,30,31</sup> All input solutions were loaded into plastic syringes (BD) and connected to the PDMS device via LDPE medical tubing (Scientific Commodities Inc.). Syringe pumps drove the flow of reagents into the device, with standard flow rates of  $2 \times 25 \mu\text{L/h}$  for two identical inner solutions of PBS (Gibco) with 1% (w/v) Tween-20 (Fisher),  $125 \mu\text{L/h}$  for the picoreactor fluorocarbon oil with the indicated concentration of dSurf (Fluigent), PicoSurf (Sphere Fluidics), Xdrop surf (Samplics), 008-FluoroSurfactant (RAN Biotechnologies), or ionic Krytox FSH (Miller-Stephenson), and  $3500\text{--}3700 \mu\text{L/h}$  for the outer sheath solution of PBS with 1% (w/v) Tween-20 and 2% (w/v) Pluronic F68 (Millipore Sigma). The PBS inner solution was labeled with  $20 \mu\text{M}$  fluorescein (Fluka).

**Microscopy.** Picoreactors were imaged with bright-field and fluorescence microscopy (Nikon). Flat-field correction, particle detection, and fluorescence quantification were performed by using custom Python scripts and the cv2/OpenCV library.

**FACS.** Picoreactors were sorted on a FACSAria II cell sorter (BD) equipped with a  $130 \mu\text{m}$  nozzle. Laser delays were manually calibrated with  $32 \mu\text{m}$  AccuCount Ultra Rainbow beads (Spherotech). Forward and side scattering voltages were manually optimized using DE samples; droplet delays were manually calibrated by sorting 50 DEs onto a glass slide at each setting and manually counting the recovered picoreactors by optical microscopy. Detector voltages were optimized by using DE samples labeled with fluorescein. Samples were loaded in the specified volume and configuration (library tube, FACS tube, FACS tube + corkscrew) as described in the figures.

## DATA AVAILABILITY

All images, CSV data files, FACS workspace files, and Autocad files are available in the Open Science Framework (OSF) at <https://osf.io/jzyve/>. Stereolithography files (.stl) for the 3D corkscrew are available on Thingiverse at <https://www.thingiverse.com/thing:6675740>, and the drawing exchange files (.dxf) are available on the OSF repository listed above.

## ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.analchem.5c01536>.

Data tables to reproduce plots of image and FACS analysis from main figures, example outputs from the automated image analysis scripts, raw output plots from FACS analysis, loading analysis with additional surfactants, and extended methods (PDF)

A literature review (XLSX)

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### Author Contributions

▲Z.Y. and S.T. contributed equally to this work. Z.Y. and S.M.T. conceived the project. Z.Y. and S.M.T. prepared all experimental samples. I.R., J.V.D., and G.V.I. designed and prepared the corkscrew mixer. S.M.T. performed all microscopy experiments. Z.Y., Y.Z., and L.N. performed all FACS analysis and sorting. P.F., G.V.I., H.T.S., J.L., and L.N. provided supervision. Z.Y., S.T., and P.F. prepared all figures and drafted the manuscript. S.M.T., P.F., and J.L. provided funding. All authors provided manuscript edits and have given approval to the final version of the manuscript.

### Notes

The authors declare no competing financial interest.

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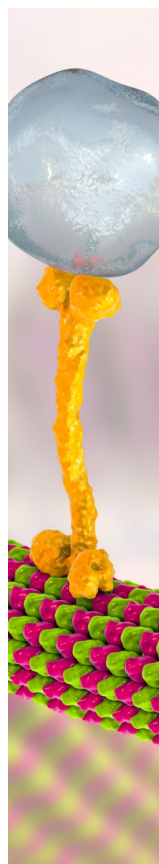
## ABBREVIATIONS

DE double emulsion or double emulsion picoreactor  
FACS fluorescence-activated cell sorting  
FADS fluorescence-activated droplet sorting  
MFI mean fluorescence intensity  
RFU relative fluorescence unit  
RT-PCR real-time polymerase chain reaction

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